



AFYA PLUS HEALTH

**A presentation on proposed medical
triage sorting chatbot**

JULY 2026

AGENDA

- 1 Business Problem
 - 2 Model Selection
 - 3 Operational Risks and Constraints
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BUSINESS PROBLEM

Afya Plus Health currently have an AI medical sorting pipeline that processes incoming patient messages, prioritises them by urgency, distinguishes high-risk emergencies from routine inquiries, and routes them appropriately. It currently has the following challenges:

- Unreliable processing due to free form text submitted by patients.
 - Hallucinations of clinical facts and conversational fluff.
 - System crashes when network degrades.
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BUSINESS IMPACT

15% LOWER PATIENT SATISFACTION

30% REDUCED PATIENT CONFIDENCE

10% DECREASE IN PROFITABILITY

GOAL

INCREASE TRIAGE ACCURACY FROM TO 98%

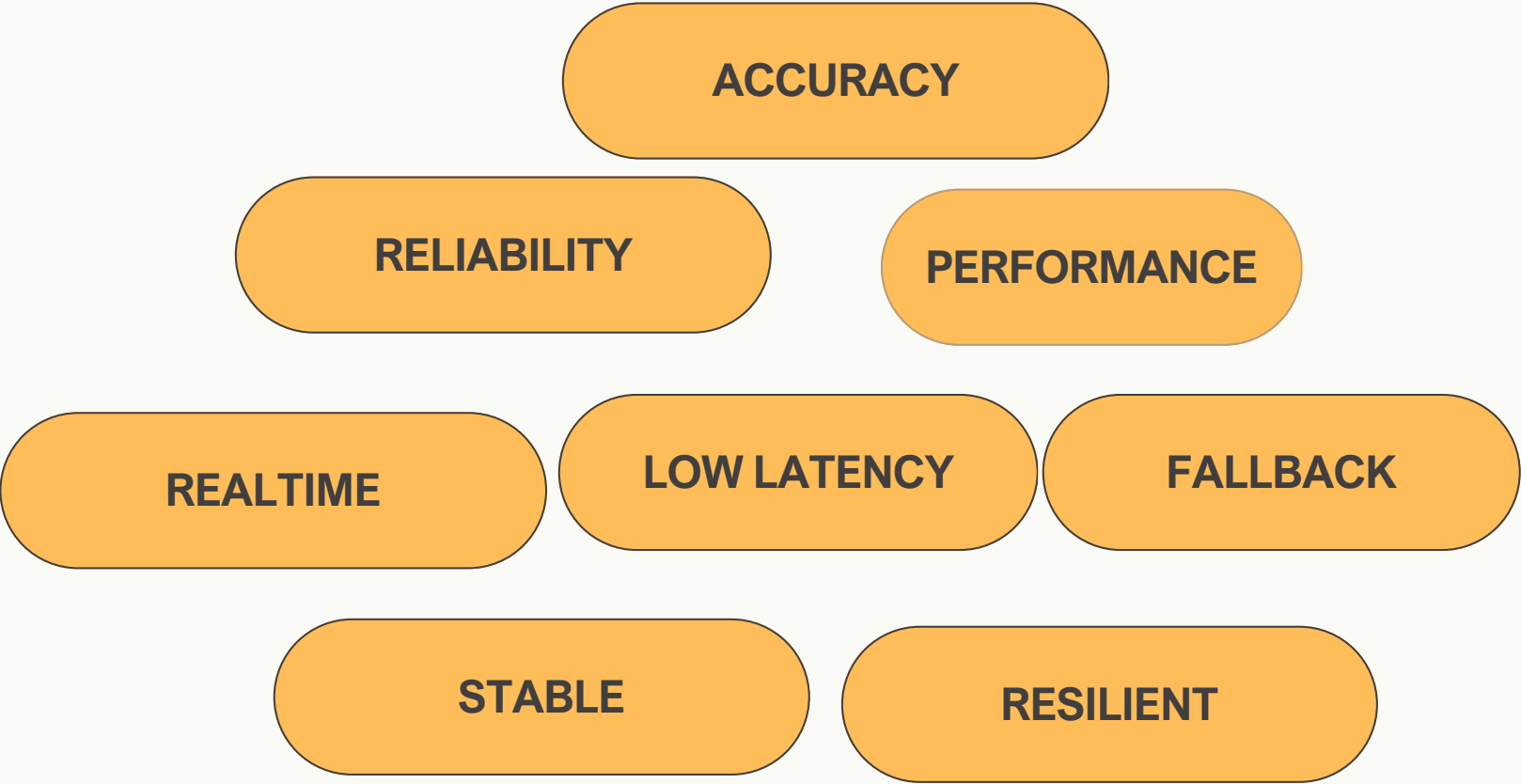
ACHIEVE 99.9% SYSTEM AVAILABILITY

RESTORE PROFITABILITY BY MORE THAN 10%

MODEL SELECTION

Given the business problem, the selected model needed to reliably transform unstructured patient messages into accurate, structured outputs while operating efficiently in a production environment. The evaluation focused on the following key criteria:

- Accuracy – Consistently identifies symptoms and classifies patient urgency correctly.
- Reliability – Produces consistent outputs and maintains stable operation.
- Performance – Delivers low-latency responses suitable for real-time triage.
- Operational Continuity – Remains available through a resilient cloud-first, local-fallback architecture.



Selection Verdict: A hybrid model strategy was selected, with the cloud model serving as the primary inference engine due to its superior accuracy and reasoning capabilities. A local model was adopted as a fallback to ensure uninterrupted service during network or cloud service failures.

RISKS & CONSTRAINTS

- Network Dependency: Cloud model performance depends on stable internet connectivity.
- Local Hardware: The local model's performance is constrained by the available CPU, memory, and storage.
- Model Accuracy: Both models may occasionally misclassify or hallucinate, requiring careful prompt engineering and validation.
- Response Format: The system relies on the AI model consistently producing valid JSON outputs for downstream processing.
- Inference Latency: Response times vary between the cloud and local models and may impact real-time triage performance.

NETWORK FAILURES

LOCAL HARDWARE

COST

DATA PRIVACY

CONSISTENCY

LATENCY
